

The z integral of two WW kernels

Where it diverges, how to regulate it, the transverse logarithm, and the splitting function

$$r \equiv y' - y, \quad \hat{r} = r/|r|, \quad d = 2 - 2\epsilon, \quad \zeta \equiv \frac{k^+}{p^+ + k^+}, \quad \bar{\zeta} = 1 - \zeta$$

The object.

$$T^{mm'}(y, y') = \int d^2 z \frac{(y - z)^m}{(y - z)^2} \frac{(y' - z)^{m'}}{(y' - z)^2}$$

The answer. It has **no ultraviolet divergence**; what diverges is the **infrared** end, $|z| \rightarrow \infty$, logarithmically. With a disc cutoff

$$T^{mm'} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'},$$

and in $d = 2 - 2\epsilon$ with $\epsilon > 0$

$$T_\epsilon^{mm'} = C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right].$$

Because $r = y' - y$ is conjugate to the measured momentum, the resulting logarithm $\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)$ is the **collinear** one, and in the two- $\mathcal{C}$ row its coefficient comes out as exactly $P_{gg}(\zeta)/2N_c$.

1. Where it diverges — and where it does not

Three regions, by power counting (n points collapsing at scale ρ give measure $\rho^{2(n-1)} \frac{d\rho}{\rho}$):

region	integrand	measure	verdict
$z \rightarrow y$	$\frac{1}{ y-z }$ (one kernel only)	$\rho d\rho$	convergent
$z \rightarrow y'$	$\frac{1}{ y'-z }$	$\rho d\rho$	convergent
$ z \rightarrow \infty$	$\frac{\hat{z}^m \hat{z}^{m'}}{ z ^2}$	$ z ^2 \frac{d z }{ z }$	log divergent

The point about the last row is that $\langle \hat{z}^m \hat{z}^{m'} \rangle = \frac{\delta^{mm'}}{2} \neq 0$: there is no angular cancellation, so the logarithm is real. And it is **infrared**, not UV. Numerically, with $y = (0.8, -0.5)$, $y' = (-0.2, 0.7)$:

cutoff R	T^{mm} (trace)	$\pi \log \frac{R^2}{r^2}$
10^2	26.1161	26.1328
10^3	40.5921	40.6004
3×10^3	47.5073	47.5032

2. Two master integrals

Write $z = z^1 + iz^2$. For a 2D vector a the combination $a^1 - ia^2$ is $\bar{a}$, so

$$\frac{a^m}{a^2} \longleftrightarrow \frac{\bar{a}}{|a|^2} = \frac{1}{a} \quad \text{and} \quad \frac{1}{\bar{a}}.$$

Everything reduces to two integrals over the disc $|z| < R$.

2.1 The holomorphic one

$$\boxed{\int_{|z|<R} \frac{d^2 z}{z - y} = -\pi \bar{y}} \quad (|y| < R)$$

Proof. Split the disc at $|z| = |y|$. For $|z| > |y|$ expand $\frac{1}{z-y} = \sum_{n \geq 0} \frac{y^n}{z^{n+1}}$; every term has a nonzero power of $e^{i\theta}$ and its angular integral vanishes. For $|z| < |y|$ expand $\frac{1}{z-y} = -\frac{1}{y} \sum_{n \geq 0} \left(\frac{z}{y}\right)^n$; only $n = 0$ survives the angular integral, leaving $-\frac{1}{y} \cdot \pi |y|^2 = -\pi \bar{y}$. $\square$ (numeric $-2.51292 - 1.57113i$ vs exact $-2.51327 - 1.57080i$)

2.2 The mixed one

$$L(a, b) = \int_{|z| < R} \frac{d^2 z}{(\bar{z} - \bar{a})(z - b)} = \pi \log \frac{R^2}{|a - b|^2}$$

Proof. Use $\frac{1}{\bar{z} - \bar{a}} = \partial_{\bar{z}} \log |z - a|^2$ and $\partial_{\bar{z}} \frac{1}{z - b} = \pi \delta^{(2)}(z - b)$, then integrate by parts:

$$L = \underbrace{\oint_{|z|=R} \log |z - a|^2 \frac{dz}{2i(z - b)}}_{\rightarrow \pi \log R^2} - \pi \log |a - b|^2. \quad \square$$

($R = 20$: numeric 16.040 vs exact 16.020; $R = 50$: 21.794 vs 21.778.)

3. The two contractions

Set $u = y - z, v = y' - z$, so $u - v = r$.

3.1 The trace

$$T^{11} + T^{22} = \text{Re} \int d^2 z \frac{1}{u \bar{v}} = \text{Re} \int \frac{d^2 z}{(z - y)(\bar{z} - \bar{y}')} = L(y', y) = \pi \log \frac{R^2}{r^2}.$$

3.2 The traceless part

$$T^{11} - T^{22} - i(T^{12} + T^{21}) = \int d^2 z \frac{1}{u \bar{v}} = \int \frac{d^2 z}{(z - y)(z - y')} = \frac{1}{y - y'} \int d^2 z \left[\frac{1}{z - y} - \frac{1}{z - y'} \right],$$

and §2.1 gives

$$= \frac{-\pi \bar{y} + \pi \bar{y}'}{y - y'} = -\pi \frac{\bar{r}}{r} = -\pi e^{-2i\phi}, \quad r = |r| e^{i\phi}.$$

(numeric $0.5700 - 3.0851i$ vs exact $0.5665 - 3.0901i$.)

3.3 Solve

Parity forbids an $\epsilon^{mm'}$ piece, and T is symmetric (swapping $m \leftrightarrow m'$ is $y \leftrightarrow y'$, i.e. $r \rightarrow -r$, which both structures are even under), so

$$T^{mm'} = A \delta^{mm'} + B \hat{r}^m \hat{r}^{m'}.$$

Matching: $\hat{r}^1 \hat{r}^1 - \hat{r}^2 \hat{r}^2 = \cos 2\phi$ and $2\hat{r}^1 \hat{r}^2 = \sin 2\phi$, so §3.2 gives $B = -\pi$; §3.1 gives $2A + B = \pi \log \frac{R^2}{r^2}$. Hence

$$T^{mm'} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}$$

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numeric T = [[24.03952, 1.54893], [1.54647, 23.47224]]
closed T = [[24.03486, 1.54505], [1.54505, 23.46834]]
antisymmetric part of numeric T: 2.5e-03 -> T IS SYMMETRIC (needed in §6)
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4. Dimensional regularisation

Take $d = 2 - 2\epsilon$ with $\epsilon > 0$: that is what damps the *large*- $|z|$ end. Feynman-parametrise, shift $z = z' + xy + \bar{x}y'$ so that $xu^2 + \bar{x}v^2 = z'^2 + \Delta$ with $\Delta = x\bar{x}r^2$, and note $u = -z' + \bar{x}r, v = -z' - xr$:

$$u^m v^{m'} \rightarrow z'^m z'^{m'} - x\bar{x} r^m r^{m'} \rightarrow \frac{\delta^{mm'}}{d} z'^2 - x\bar{x} r^m r^{m'}.$$

With the two standard Euclidean integrals

$$\int \frac{d^d l}{[l^2 + \Delta]^2} = \pi^{d/2} \Gamma(2 - \frac{d}{2}) \Delta^{\frac{d}{2}-2}, \quad \int \frac{d^d l l^2}{[l^2 + \Delta]^2} = \pi^{d/2} \frac{d}{2} \Gamma(1 - \frac{d}{2}) \Delta^{\frac{d}{2}-1},$$

both x integrals reduce to the same Euler beta, $\int_0^1 (x\bar{x})^{-\epsilon} dx = \frac{\Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)}$, and

$$T_\epsilon^{mm'} = C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad C(\epsilon) = \pi^{1-\epsilon} \frac{\Gamma(1+\epsilon) \Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)} = \pi e^{-\epsilon(\log \pi + \gamma_E - 2)}.$$

Cross-check on the trace. Contract with $\delta^{mm} = d$ and compare with the independent route through the massless bubble: write $u \cdot v = \frac{1}{2}(u^2 + v^2 - r^2)$, drop the two scaleless tadpoles, and use $\int \frac{d^d z}{(z-y)^{2a}(z-y')^{2b}} = \pi^{d/2} \frac{\Gamma(a+b-\frac{d}{2}) \Gamma(\frac{d}{2}-a) \Gamma(\frac{d}{2}-b)}{\Gamma(a) \Gamma(b) \Gamma(d-a-b)} (r^2)^{\frac{d}{2}-a-b}$. The two agree identically (symbolic difference 0).

The pole is infrared. $\epsilon > 0$ means $d < 2$, which converges at large $|z|$ — §1. Be careful here: the bubble route *does* produce $\frac{1}{\epsilon_{UV}}$ poles in intermediate steps (the scaleless tadpole is $\pi[\frac{1}{\epsilon_{IR}} - \frac{1}{\epsilon_{UV}}]$, zero only because two divergences are set equal, and $G(1,1)$ is genuinely log UV divergent at $z \rightarrow y$). They cancel between the two. The surviving pole is $\frac{1}{\epsilon_{IR}}$, so it is absorbed into the gluon distribution, **not** into α_s .

5. The transverse logarithm

Attach the $\overline{\text{MS}}$ factor $(\frac{e^{\gamma_E}}{4\pi})^\epsilon$ that rides on $g^2 \mu^{2\epsilon}$:

$$\left(\frac{e^{\gamma_E}}{4\pi}\right)^\epsilon C(\epsilon) (r^2 \mu^2)^{-\epsilon} \frac{1}{\epsilon} = \pi \left[\frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu^2 r^2) \right] \equiv \pi [\mathcal{L} + 2],$$

$$\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2), \quad r = y' - y.$$

Matching against the cutoff form of §3.3 fixes the dictionary, $\frac{\pi}{2}(\log \frac{R^2}{r^2} + 1) = \frac{\pi}{2}[\mathcal{L} + 2]$, i.e.

$$R = \frac{e^{1/2}}{2\pi\mu}.$$

Why this is the collinear logarithm. In the row, the surviving phase is $e^{-ik \cdot r}$ — $r = y' - y$ is conjugate to the *measured* transverse momentum. So $\mathcal{L}$ is the logarithm between the factorisation scale μ and the hard transverse scale r^{-1} , which the remaining y, y' Fourier transform ties to $k_\perp$. A splitting function times a logarithm of that ratio is exactly the DGLAP structure. (R is the *outer* radius, so μ here is a factorisation scale, not a renormalisation scale — do not run α_s with it.)

6. Feeding it back: the $2/k^+$ term dies

In the two- $\mathfrak{C}$ row the two vertices contract to three structures. Writing them in the splitting variable $\zeta = \frac{k^+}{S}$, $S = p^+ + k^+$ — the $\mathfrak{C}$ vertex splits **one** gluon of S into the measured k^+ and the unobserved p^+ , so $a = \frac{S}{p^+} = \frac{1}{\zeta}$ and $b = \frac{S}{k^+} = \frac{1}{\zeta}$ —

$$c_1 = d_\perp - \frac{2}{\zeta} - \frac{2}{\zeta} (\delta_{km} \delta_{k'm'}), \quad c_2 = \frac{1}{\zeta^2} + \frac{1}{\zeta^2} (\delta_{mm'} \delta_{kk'}), \quad c_3 = \frac{2}{\zeta \zeta} (\delta_{km'} \delta_{k'm}).$$

Because $T^{mm'}$ is symmetric (§3.3), $\delta_{km'} \delta_{k'm}$ and $\delta_{km} \delta_{k'm'}$ give the *same* contraction, so only $c_1 + c_3$ acts:

$$c_1 + c_3 = d_\perp + \frac{2}{\zeta \zeta} [1 - \zeta - \bar{\zeta}] = d_\perp.$$

The $\frac{2}{k^+}$ term of the bracket cancels, and in this variable the reason is plain: it is **the momentum sum rule** $\zeta + \bar{\zeta} = 1$.

Keeping $d_\perp = 2 - 2\epsilon$ consistently, so that $\frac{d_\perp}{2\epsilon} = \frac{1}{\epsilon} - 1$ and $\frac{\text{tr } T}{C(\epsilon)(r^2\mu^2)^{-\epsilon}} = \frac{d}{2\epsilon} - 1 = \frac{1}{\epsilon} - 2$, and using $\frac{C(\epsilon)(\dots)^{-\epsilon}}{\epsilon} \rightarrow \pi[\mathcal{L} + 2]$ while $C(\epsilon)(\dots)^{-\epsilon} \rightarrow \pi$, the whole z -integrated structure collapses to

$$\frac{\pi}{k^{+2}} \left\{ C_{\text{UV}}(\zeta) \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}$$

$\mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}$, $\mathcal{R} = \frac{[r \cdot (x-y)][r \cdot (x'-y')]}{r^2 (x-y)^2 (x'-y')^2}$. Symbolic difference from the direct ϵ -expansion: exactly 0. The $\frac{1}{\zeta^2}$ that appears on the way is absorbed by $S^2 = k^{+2}/\zeta^2$.

7. That coefficient is the splitting function

$$C_{\text{UV}}(\zeta) = \frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta \bar{\zeta} = \zeta \bar{\zeta} + \frac{1}{\zeta \bar{\zeta}} - 2 = \frac{P_{gg}(\zeta)}{2N_c}, \quad P_{gg}(\zeta) = 2N_c \left[\frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta \bar{\zeta} \right].$$

The colour reduction of the row supplies exactly one factor N_c ($U^{d'b}(z)U^{db}(z) = \delta^{d'd}$, then $f^{c'da}f^{cda} = N_c \delta^{c'c}$), so the two combine into $N_c C_{\text{UV}} = \frac{P_{gg}}{2}$ with nothing left over. And the non-logarithmic remainder carries $\zeta \bar{\zeta}$ — the **regular** piece of P_{gg} , the only term with no endpoint pole.

7.1 The convolution

At fixed k^+ , $S = k^+/\zeta$, so $dp^+ = \frac{k^+}{\zeta^2} d\zeta$ and $\frac{d\zeta}{\zeta^2} = \frac{1}{\zeta} \frac{dS}{S}$ — the usual DGLAP measure:

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{ec}} &= \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{1}{2k^+} \int_{k^+/\vee}^{k^+/(k^++\Lambda)} \frac{d\zeta}{\zeta^2} \left\{ \frac{P_{gg}(\zeta)}{2} \mathcal{L} \mathcal{X} + N_c \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\} \\ &\times \int_{x, x', y, y'} e^{-ik \cdot (y' - y)} [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x) \end{aligned}$$

The collinear counterterm that removes the $\frac{1}{\epsilon}$ is the matching $-\frac{1}{\epsilon} \frac{P_{gg}}{2} \otimes (\text{LO})$.

7.2 The two endpoints

The ζ integral is elementary, $\int \frac{d\zeta}{\zeta^2} C_{\text{UV}}(\zeta) = F(\zeta)$ with

$$F(\zeta) = 2 \log \zeta - \log(1 - \zeta) - \zeta + \frac{1}{\zeta} - \frac{1}{2\zeta^2} \quad (F' = C_{\text{UV}}/\zeta^2 \checkmark),$$

- $\zeta \rightarrow 1$ is $p^+ \rightarrow 0$: $-\log(1 - \zeta) \rightarrow \log \frac{k^+}{\Lambda}$ — the **rapidity** logarithm, coefficient exactly 1. This is what goes to JIMWLK, and what $\left[\frac{1}{p^+}\right]_+$ isolates.
- $\zeta \rightarrow 0$ is $p^+ \rightarrow \vee$: $\frac{1}{\zeta^3}$ gives $\frac{\vee^2}{2k^{+2}}$ — the kinematic end, where the eikonal treatment must be cut.
- Restricting to $p^+ \leq k^+$ (i.e. $\zeta \geq \frac{1}{2}$) removes the second problem entirely:

$$\int_{1/2}^{\zeta_+} \frac{d\zeta}{\zeta^2} C_{\text{UV}}(\zeta) = \log \frac{2k^+}{\Lambda} \quad (\text{checked: } 14.508658 \text{ vs } 14.508658 \text{ at } k^+ = 1, \Lambda = 10^{-6}).$$

And the familiar sum rule sits in the same function: $\int_0^1 [\zeta \bar{\zeta} + \frac{1}{\zeta \bar{\zeta}} - 2] d\zeta = \frac{1}{6} - 2 + (\text{endpoint logs}) = 2\ell_k - \frac{11}{6}$, which times $2N_c$ is $4N_c \ell_k - \frac{11N_c}{3} = 4N_c \ell_k - b_0$.

zint/src/note_zint.py (§1–5), note_dglap.py (§5–7) — every number and identity above.